CIVIX 2.0 — Project Master Reference Document

Document Status: Authoritative Project Master Reference & Workstation Blueprint
Audience: Senior Engineering Leadership, SIH Evaluators, System Architects, Lead Investigators
Authority Rule: Derived directly from `docs/00_CIVIX_CURRENT_STATE.md` and repository code inspection.

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1. Executive Summary & Core Mission

CIVIX 2.0 is a state-of-the-art **Investigative Intelligence Workstation** engineered to solve the critical challenge of **fragmented, multi-modal investigative data** in complex law enforcement, counter-fraud, and intelligence operations.

In modern criminal investigations, data arrives in disconnected, heterogeneous silos: - **Telecom Records**: Call Detail Records (CDRs), IP Detail Records (IPDRs), IMEI/IMSI pairing logs, cell tower coverage maps. - **Financial Networks**: Bank ledger transactions, UPI transfers, credit card logs, high-value cash deposits, shell account networks. - **Computer Vision & Video**: CCTV video feeds, Automated License Plate Recognition (ANPR), SORT vehicle tracking logs, facial embeddings. - **Geospatial Telemetry**: Cell tower GIS centroids, vehicle movement vectors, PostGIS location points, spatiotemporal co-location events. - **Unstructured Intelligence**: Witness statements, PDF FIR reports, police case diaries, forensic reports, uploaded evidence vaults.

CIVIX 2.0 unifies these fragmented data points into a single, cohesive, **evidence-backed, security-enforced investigative workstation**. It bridges relational transactions with graph-oriented multi-hop link analysis while strictly maintaining **epistemic provenance**, **Row-Level Security (RLS)**, and **bitemporal court admissibility**.

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2. Complete Technology Stack Inventory

Component Layer	Technology / Framework	Version / Package	Purpose & Implementation Details
Frontend Framework	React 18 (TypeScript), Vite 5	React 18.2, Vite 5.0	High-performance Single Page Application (SPA) with strict type safety, modular routing, and reactive state management.
Graph Visualization Engine	Cytoscape.js	`cytoscape` 3.28, `cytoscape-fcose`	Hardware-accelerated canvas/SVG multi-hop graph rendering, fCoSE layout physics, dynamic node clustering, and edge filtering.
GIS & Spatial Mapping	Leaflet, React-Leaflet	Leaflet 1.9, OpenStreetMap	Interactive GIS map canvas rendering PostGIS spatial centroids, cell tower radii, vehicle trajectory vectors, and spatiotemporal timeline scrubbing.
UI Component System	TailwindCSS, Lucide Icons	Tailwind 3.4, Lucide React	Modern dark-mode glassmorphic user interface, high-contrast visual cues, responsive grid layouts, and custom CSS variables.
Asynchronous API Server	Python 3.12, FastAPI	FastAPI 0.109, Uvicorn 0.27	Asynchronous REST API server handling JWT authentication, request validation (Pydantic v2), OpenAPI documentation, and route dispatching.
Primary System of Record	PostgreSQL 16 + PostGIS	PostgreSQL 16.1, PostGIS 3.4	Authoritative transactional data store, bitemporal schema triggers, strict Row-Level Security (RLS), and PostGIS spatial indexing (`GEOMETRY`).
Graph Database Projection	Neo4j 5 Enterprise / Community	Neo4j 5.15, Cypher 5	Read-optimized multi-hop graph traversal engine executing 1–5 hop Cypher queries, path analysis, and graph centrality calculations.
CDC / Event Synchronization	PostgreSQL Outbox + Async Python Worker	Transactional Outbox Pattern	Asynchronous, idempotent event synchronization queue (`civix.outbox`) ensuring PostgreSQL mutations project to Neo4j without distributed transactions.
Behavioral Anomaly ML	XGBoost 2.0, scikit-learn	XGBoost 2.0.3, NumPy 1.26	Supervised 70-feature behavioral anomaly classification model (`xgboost_behavioral_v1.bin`) producing candidate risk scores [0.00–1.00].
Computer Vision (CV)	YOLOv8, SORT Tracker, EasyOCR	Ultralytics YOLOv8, PyTorch	Real-time vehicle license plate detection, Kalman-filter SORT multi-object tracking (`track_id`), and optical character recognition (ANPR).
Multi-Modal Gemini NLP	Google Gemini Flash 1.5, Groq LLaMA 3	`google-generativeai`	Automated evidence document parsing, unstructured text entity extraction (people, vehicles, bank accounts), and structured triple mapping.
Database ORM / Drivers	SQLAlchemy 2.0, asyncpg, Neo4j Driver	SQLAlchemy 2.0 (AsyncIO)	Asynchronous connection pooling, PostgreSQL session lifecycle management, and Cypher driver execution.

3. Workstation Pages & User Interface Deep Dive

CIVIX 2.0 features 13 specialized frontend workspace modules (`frontend/src/pages/`):

1. `CaseWorkspacePage.tsx` **(Case Management Engine)**:
2. Central hub for managing cases, assigning investigator permissions, configuring case ACLs (`READ` , `WRITE` , `ADMIN`), viewing entity dossiers, and tracking investigation milestones.
3. `InvestigativeGraphPage.tsx` **(Multi-Hop Graph Explorer)**:
4. Interactive multi-hop graph canvas (`GraphExplorer.tsx` , `GraphCanvas.tsx`) supporting 1H–5H expansions, node focus mode, path analysis panel (`PathAnalysisPanel.tsx`), investigator proposal creation (`ProposalDrawer.tsx`), relationship inspector (`RelationshipInspector.tsx`), and epistemic legend controls (`EpistemicLegend.tsx`).
5. `EntityDossierPage.tsx` **(Complete Entity Profiler)**:
6. Deep-dive dossier for any entity (`PERSON` , `VEHICLE` , `TELECOM` , `LOCATION`). Displays bitemporal attribute history, connected assertions, timeline of events, XGBoost risk scores, and linked evidence documents.
7. `SpatialIntelligencePage.tsx` **(GIS & Timeline Scrubber)**:
8. Interactive PostGIS spatial map rendering cell tower centroids, movement vectors, CCTV camera locations, and a time-range scrubber to analyze entity movement patterns over time.
9. `TelecomIntelligencePage.tsx` **(CDR & Cell Tower Analytics)**:
10. Call Detail Record (CDR) matrix aggregator (`TelecomMatrix.tsx`), co-location analysis across cell towers, subscriber call frequency heatmaps, IMEI/IMSI switching detectors, and night call volume charts.
11. `CCTVCommandCenterPage.tsx` **(Live Video & ANPR Tracker)**:
12. Multi-camera video feed monitoring dashboard (25 Delhi NCR streams), real-time ANPR license plate detection logs, SORT tracking ID persistence (`track_id`), and camera location map pins.
13. `BiometricIntelligencePage.tsx` **(Biometric Matching Interface)**:
14. Facial feature extraction, fingerprint pattern matching, biometric similarity scoring UI, and gallery comparison view.
15. `EvidencePage.tsx` **(Multi-Modal Evidence Vault)**:
16. Central document upload vault (`EvidenceVault.tsx`), PDF viewer, Gemini Flash 1.5 extracted entity tags, and court evidence chain-of-custody verification logs.
17. `VisualAnalysisPage.tsx` **(Visual Analytics Workspace)**:
18. Deep visual analytics workspace displaying bounding box overlays, object class distributions, and spatiotemporal visual clusters.
19. `CommandCenterPage.tsx` **(Executive Operational Dashboard)**:
    - High-level executive dashboard showing active case counts, pending supervisor assertion review counters, system telemetry indicators, and high-risk candidate alerts.
20. `SearchPage.tsx` **(Ask CIVIX Natural Language Search)**:
    - Natural language query bar powered by Gemini Flash 1.5, converting plain text questions ("*Show all calls between Suspect A and Suspect B in Jaipur during midnight*") into SQL/Cypher queries.
21. `FoundationTestPage.tsx` **(System Diagnostic Console)**:
    - Internal engineering diagnostic panel verifying backend API availability, database connection health, and Neo4j connectivity.
22. `CasesPage.tsx` **(Case Directory)**:
    - Master list of all cases accessible to the logged-in investigator, filtered according to PostgreSQL Row-Level Security (RLS) policies.

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4. Domain Ontology & Entity-Relation Model

CIVIX 2.0 uses a standardized, extensible domain ontology (`civix` PostgreSQL schema):

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|                              CIVIX 2.0 DOMAIN ONTOLOGY                            |
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| CORE ENTITIES (`civix.entity`):                                                    |
|   - PERSON            : Human target, suspect, victim, associate                  |
|   - VEHICLE           : Automobile, motorcycle, truck (Plate #, VIN, Make, Model) |
|   - TELECOM           : MSISDN (Phone #), IMSI, IMEI, Cell Tower Sector ID        |
|   - FINANCIAL         : Bank Account, UPI VPA, Credit Card, Wallet ID             |
|   - LOCATION          : GIS Point, Address, Landmark, Centroid (SRID 4326)        |
|   - ORGANIZATION      : Company, Shell Firm, Gang, Bank, Agency                   |
|   - EVIDENCE          : PDF Document, Image, Video Clip, Audio Recording          |
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| SPATIOTEMPORAL EVENTS (`civix.event` & `civix.event_participant`):                 |
|   - CALL / MESSAGE    : Telecom communication event (Roles: CALLER, CALLEE, TOWER) |
|   - TRANSACTION       : Financial transfer event (Roles: SENDER, RECEIVER)        |
|   - CCTV_SIGHTING     : Video ANPR capture (Roles: SUBJECT, CAMERA_LOCATION)     |
|   - DEVICE_PING       : IPDR / Data session ping (Roles: DEVICE, CELL_TOWER)      |
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| CONTROLLED PREDICATES (`civix.assertion` & `predicate_enum`):                     |
|   CALLED, MESSAGED, CO_LOCATED, REGISTERED_TO, OWNED_BY, DRIVER_OF, MEMBER_OF,   |
|   EMPLOYED_BY, ASSOCIATED_WITH, KNOWN_ASSOCIATE_OF, OBSERVED_AT, TRANSFERRED_TO,  |
|   RECEIVED_FROM, ALIAS_OF, LIVES_AT, WORKS_AT, FINANCES, HAS_SIM, USES_DEVICE,    |
|   TRANSACTED_WITH, PRESENT_AT                                                     |
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5. Core Architectural Principles & Epistemic Separation

CIVIX 2.0 operates under strict engineering principles to eliminate AI hallucination, prevent unauthorized data leakage, and guarantee court-admissible evidence.

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|                              EPISTEMIC SEPARATION MODEL                           |
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| 1. OBSERVED FACTS         : Physical ground truth (CDRs, CCTV captures, Bank TXs) |
| 2. AI / ML INFERENCES    : System suggestions (XGBoost score, OCR, Gemini NLP)    |
| 3. HUMAN HYPOTHESES       : Investigator theories & stance support               |
| 4. INVESTIGATOR PROPOSAL  : PROPOSED relationship (Pending Supervisor Approval)   |
| 5. CONFIRMED FACT         : ACCEPTED_BY_SUPERVISOR → Neo4j INVESTIGATOR_ASSERTED |
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The 5 Epistemic Invariants

1. **INV-01 (Fact Integrity)**: Observed ground truth (telecom logs, bank transfers, CCTV frames) is immutable and can never be mutated or overwritten by AI predictions or investigator assertions.
2. **INV-08 (No Autonomous AI/Investigator Confirmation)**: Neither AI inferences nor raw investigator proposals can self-confirm or directly alter the primary graph projection without supervisor approval.
3. **INV-18 (Controlled Predicates)**: All relationship types must conform to strict PostgreSQL `predicate_enum` definitions. Free-text strings are strictly prohibited.
4. **ADR-REM-02 (Unconfirmed Proposal Isolation)**: Relationships with status `PROPOSED` reside exclusively in PostgreSQL and are **NEVER** projected into the Neo4j graph traversal layer.
5. **ADR-REM-03 (Supervisor-Approved Projection)**: Only upon explicit supervisor review (`ACCEPTED_BY_SUPERVISOR`) does an outbox event trigger projection of an `INVESTIGATOR_ASSERTED` edge into Neo4j.

6. Major Limitations & Operational Gaps

1. **Government System Integrations**: CIVIX 2.0 does **NOT** connect to live government databases (CCTNS, NCRB, Aadhaar/UIDAI, telecom operator networks, or live police CCTV networks). All inputs use local uploads or synthetic test datasets.
2. **Neo4j Polling CDC Lag**: Graph projections rely on a periodic Python outbox worker (`worker/outbox_worker.py`). A 1–2 second synchronization delay exists between PostgreSQL commit and Neo4j edge availability.
3. **Biometric Face Search Backend**: The frontend biometric facial matching UI utilizes synthetic mock distance scores; live vector search (e.g. Pgvector/Milvus) is scheduled for Phase 2.